



Deresegn POS

Mobile API User Documentation

API Guide & Gateway Specifications

Document Version: 1.0

Application Version: 1.0.3+4

Date: August 5, 2026

Classification: Confidential

Prepared by: Micro Sun & Solution PLC

Email: amanuielt@mssethiopia.com

Phone: +251 911 058 179

Deresegn POS - Mobile API User Documentation (API Guide)

1. Introduction & API Overview

1.1 Purpose

This document serves as the official API User Documentation (API Guide) for the Deresegn POS platform. It specifies how mobile client applications and external API consumers interact with the Deresegn Backend API (`https://api.deresegn.com`).

The API acts as a secure, authenticated gateway and fiscal proxy between the mobile application and the Ethiopian Ministry of Revenues (MoR) e-invoicing platform.

1.2 Regulatory & Technical Specification References

The API endpoints, encryption parameters, payload schemas, and fiscal verification workflows documented herein are built in strict compliance with the non-public technical specifications issued directly by the Ministry of Revenues (MoR) to registered e-invoicing solution providers:

- Reference Standard: Ministry of Revenues (MoR) Electronic Invoicing System Integration & API Specification
- Issuing Authority: Ministry of Revenues, Federal Democratic Republic of Ethiopia
- Classification: Restricted / Vendor Provisioned (Confidential)
- Compliance Status: Fully compliant with MoR RSA-SHA512 digital signature, e-Invoice IRN schema, sales receipt, and withholding tax registration protocols.
- Attachment: Copies of the official MoR Specification documents are available upon request and included under the `References` folder in the vendor submission archive.

1.3 Base URL & Environment

- Production Base URL: `https://api.deresegn.com`
- Protocol: HTTPS (TLS 1.2+ mandatory)
- Data Format: `application/json` (except certificate/key upload endpoints which use `multipart/form-data`)

1.4 Common HTTP Headers

Header Name	Requirement	Description
`Content-Type`	Mandatory	`application/json`
`Accept`	Mandatory	`application/json`
`Authorization`	Conditional	`Bearer <JWT_TOKEN>` (Company or Branch token)
`Branch-Id`	Conditional	Numeric ID of the authenticated branch
`Mor-Token`	Conditional	`Bearer <MOR_ACCESS_TOKEN>` for MoR passthrough routes

2. Authentication & Authorization Flow

The API utilizes a dual-tier JWT authentication model:

- Company Owner Auth: Access company setup and branch creation endpoints.
- Branch Auth: Access daily point-of-sale operational endpoints (invoicing, receipts, catalog).

API Interaction Architecture Deresegn POS

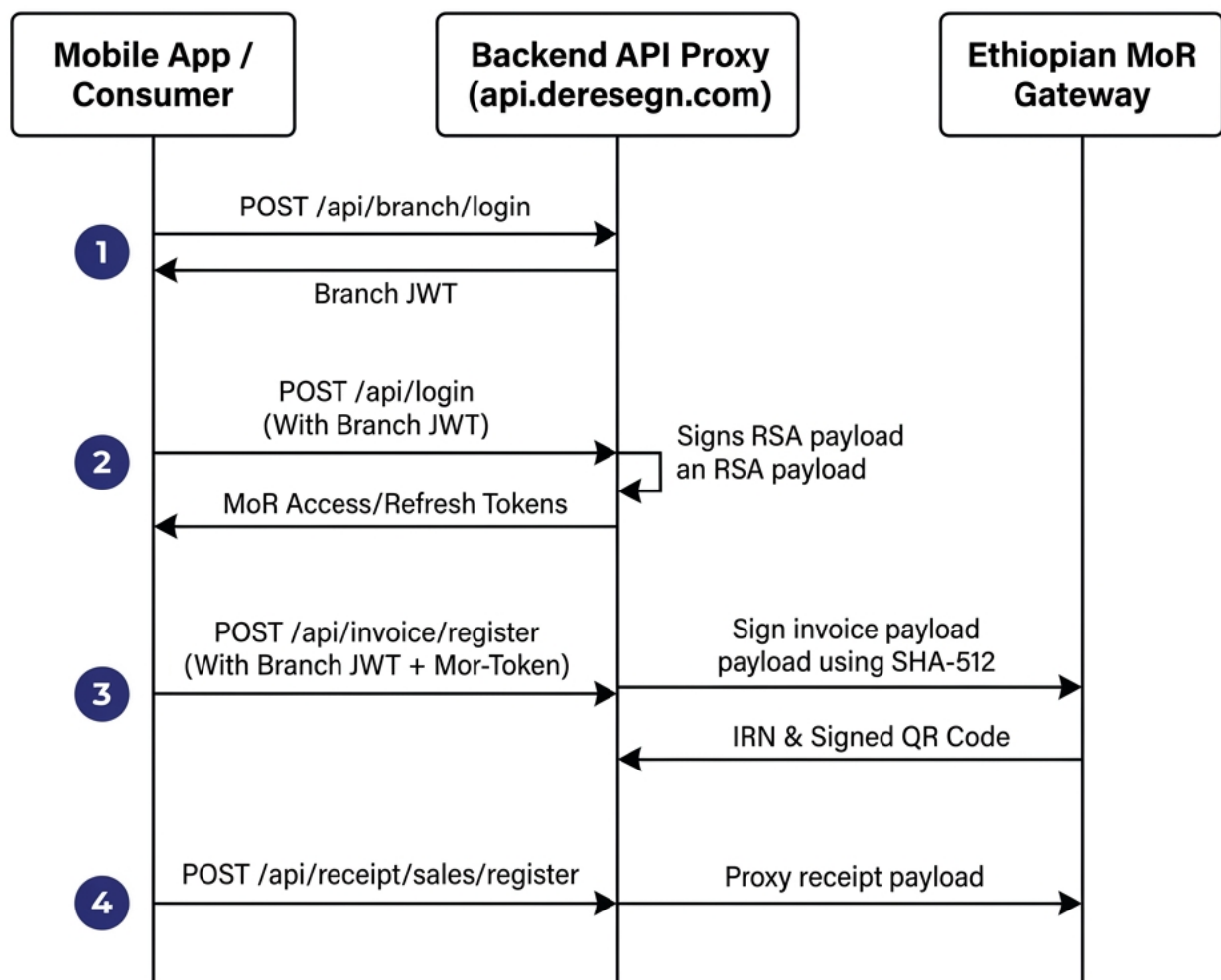


Figure: Mobile API Interaction & Gateway Architecture



3. API Endpoints & Complete Payload Specifications

3.1 Authentication Endpoints

3.1.1 Company Owner Registration

- Endpoint: `POST /api/company/register`
- Auth: Public
- Request Body:

```
{
  "name": "Acme Trading PLC",
  "tin": "0000037187",
  "phone": "+251911058179",
  "email": "owner@acmetrading.et",
  "password": "SecurePassword123!"
}
```

- Response (201 Created):

```
{
  "status": "success",
  "message": "Company registered successfully",
  "access_token": "eyJhbGciOiJIUzI1Ni...",
  "token_type": "bearer",
  "expires_in": 3600
}
```

3.1.2 Company Owner Login

- Endpoint: `POST /api/company/login`
- Auth: Public
- Request Body:

```
{
  "phone": "+251911058179",
  "password": "SecurePassword123!",
  "company_id": 1
}
```

3.1.3 Branch Operator Login

- Endpoint: `POST /api/branch/login`
- Auth: Public
- Request Body:

```
{
  "tin": "0000037187",
  "password": "BranchPassword123"
}
```



}

3.2 Fiscal Invoice Registration (`POST /api/invoice/register`)

Regulatory MoR Specification Reference:

The `InvoiceRegisterRequest` JSON payload schema below strictly adheres to the restricted, non-public Ministry of Revenues (MoR) e-Invoicing Technical Specification provided to registered vendors. All 10 root nodes (`DocumentDetails`, `TransactionType`, `SourceSystem`, `SellerDetails`, `BuyerDetails`, `ItemList`, `ValueDetails`, `PaymentDetails`, `ReferenceDetails`, `Version`) match the exact structure and field names mandated by the MoR e-invoicing gateway.

- Endpoint: `POST /api/invoice/register`
- Auth: Required (`Authorization: Bearer <Branch\_JWT>`, `Branch-Id: <id>`, `Mor-Token: Bearer <MoR\_Token>`)
- Full Production Request Payload:

```
{
  "DocumentDetails": {
    "DocumentNumber": "10042",
    "Type": "INV",
    "Reason": "Sales Invoice",
    "Date": "05-08-2026T14:00:00"
  },
  "TransactionType": "B2C",
  "SourceSystem": {
    "SystemType": "POS",
    "CashierName": "Kidist A.",
    "SystemNumber": "POS-01",
    "InvoiceCounter": 10042,
    "SalesPersonName": "Kidist A."
  },
  "SellerDetails": {
    "Tin": "0000037187",
    "LegalName": "Micro Sun & Solution PLC",
    "City": "Addis Ababa",
    "Wereda": "13",
    "Region": "1",
    "Email": "amanuielt@mssethiopia.com",
    "Phone": "+251911058179",
    "Country": "1",
    "TradeName": "Deresegn POS",
    "VatNumber": "VAT-0000037187"
  },
  "BuyerDetails": {
    "LegalName": "Selamawit Bekele",
    "IdType": "KID",
    "HouseNumber": "",
    "IdNumber": "",
    "Tin": "0011223344",
    "Email": "",
    "Phone": "",
    "City": "Addis Ababa",
    "Region": "",
    "Country": ""
  }
}
```



```
"Kebele": "",
"Wereda": "",
"VatNumber": ""
},
"ItemList": [
{
  "LineNumber": 1,
  "NatureOfSupplies": "goods",
  "UnitPrice": "150.00",
  "TotalLineAmount": "172.50",
  "PreTaxValue": "150.00",
  "Unit": "PCS",
  "TaxCode": "VAT15",
  "TaxAmount": "22.50",
  "Quantity": "1.00",
  "Discount": "0.00",
  "ExciseTaxValue": "0.00",
  "ProductDescription": "Wireless Optical Mouse",
  "ItemCode": "ITEM-1"
}
],
"ValueDetails": {
```

- Response (200 OK):

```
{
  "status": "success",
  "data": {
    "irn": "IRN-2026-0805-00142",
    "documentNumber": "10042",
    "signedQrCode": "https://mor.gov.et/verify/qr?data=...",
    "registrationTime": "2026-08-05T14:00:00Z"
  }
}
```

3.2.1 Cancel Fiscal Invoice (`POST /api/cancel/invoice`)

- Endpoint: `POST /api/cancel/invoice`
- Auth: Required (Authorization: Bearer <Branch_JWT>)
- Full Payload:

```
{
  "Irn": "IRN-2026-0805-00142",
  "ReasonCode": "CUSTOMER_CANCEL"
}
```

3.2.2 Verify Fiscal Invoice (`POST /api/irn/verify`)

- Endpoint: `POST /api/irn/verify`
- Auth: Required (Authorization: Bearer <Branch_JWT>)
- Full Payload:



```
{
  "Irn": "IRN-2026-0805-00142"
}
```

3.3 Sales Receipt Registration (`POST /api/receipt/sales`)

Regulatory MoR Specification Reference:

The `ReceiptRegisterRequest` JSON payload schema below strictly adheres to the non-public Ministry of Revenues (MoR) Sales Receipt Registration Specification. Key structures including `Invoices` array mapping to IRN and `TransactionDetails` conform to official fiscal collection compliance.

- Endpoint: `POST /api/receipt/sales`
- Auth: Required (`Authorization: Bearer <Branch\_JWT>`, `Mor-Token: Bearer <MoR\_Token>`)
- Full Production Request Payload:

```
{
  "ReceiptNumber": "1785927533938",
  "ReceiptType": "Sales Receipts",
  "Reason": "Payment for goods purchased",
  "ReceiptDate": "2026-08-05T14:00:00.000Z",
  "ReceiptCounter": "1",
  "ManualReceiptNumber": null,
  "SourceSystemType": "POS",
  "SourceSystemNumber": "POS-01",
  "ReceiptCurrency": "ETB",
  "ExchangeRate": null,
  "CollectedAmount": "172.50",
  "SellerTIN": "0000037187",
  "Invoices": [
    {
      "InvoiceIRN": "IRN-2026-0805-00142",
      "PaymentCoverage": "FULL",
      "InvoicePaidAmount": "172.50",
      "TotalAmount": "172.50"
    }
  ],
  "TransactionDetails": {
    "ModeOfPayment": "CASH",
    "DocumentNumber": 10042,
    "CollectorName": "Kidist A.",
    "TransactionNumber": "TXN-998877",
    "PaymentServiceProvider": "Telebirr"
  }
}
```

- Response (200 OK):

```
{
  "status": "success",
  "data": {
    "id": 842,
    "mor_id": "MOR-REC-99120",
  }
}
```



```
{
  "status": "REGISTERED",
  "rrn": "RRN-2026-0805-99881",
  "invoice_irn": "IRN-2026-0805-00142",
  "receipt_number": "1785927533938",
  "qr": "https://mor.gov.et/receipt/qr?data=...",
  "is_success": true
}
```

3.4 Withholding Receipt Registration (`POST /api/receipt/withholding`)

Regulatory MoR Specification Reference:

The `WithholdingReceiptRequest` JSON payload schema below strictly complies with the non-public Ministry of Revenues (MoR) Transaction Withholding Tax (TWHT) Specification, embedding `InvoiceDetail` and `WithholdDetail` nodes.

- Endpoint: `POST /api/receipt/withholding`
- Auth: Required (`Authorization: Bearer <Branch\_JWT>`, `Mor-Token: Bearer <MoR\_Token>`)
- Full Production Request Payload:

```
{
  "ReceiptNumber": "WHT-1785927533938",
  "Reason": "Withhold for goods purchased",
  "ReceiptCounter": "1",
  "ManualReceiptNumber": "92753",
  "SourceSystemType": "POS",
  "SourceSystemNumber": "POS-01",
  "BuyerTIN": "0011223344",
  "InvoiceDetail": {
    "InvoiceIRN": "IRN-2026-0805-00142",
    "Currency": "ETB"
  },
  "WithholdDetail": {
    "Type": "TWHT",
    "PreTaxAmount": 150.00,
    "WithholdingAmount": 3.00
  }
}
```

- Response (200 OK):

```
{
  "status": "success",
  "message": "Withholding receipt registered successfully"
}
```

3.5 Supply / Product Catalog Management (`/api/supplies`)

- Add Supply (`POST /api/supplies`):

```
{
  "item_code": "ITEM-101",
  "product_description": "Barcoded Scanner Unit",
}
```




```
"nature_of_supplies": "goods",
"unit_price": 2500.00,
"unit": "PCS",
"tax_code": "VAT15",
"discount": 0.00,
"excise_tax_rate": 0.00,
"is_excise_taxable": false
}
```

4. Error Handling & HTTP Status Codes

4.1 Status Code Index

Code	Description	Handling Recommendation
`200 OK`	Success	Process returned payload
`201 Created`	Resource created	Store newly created resource data
`400 Bad Request`	Invalid payload or missing parameters	Check request body format
`401 Unauthorized`	Expired/invalid JWT or MoR token	If MoR error: refresh token; if Branch error: redirect to login
`403 Forbidden`	Access denied for requested branch	Verify branch ownership
`406 Not Acceptable`	Document number mismatch	Auto-retry with expected document number returned by MoR
`422 Unprocessable`	Validation rules failed	Display error messages to user
`500 Server Error`	Backend or MoR Gateway error	Queue request locally (if offline-capable) and retry

5. Document Number Auto-Correction (406 Protocol)

When registering invoices, if out-of-sync document sequence numbers occur:

- MoR Gateway returns `HTTP 406 Not Acceptable` with payload containing `expectedDocumentNumber`.
- Mobile API Client intercepts response.
- Client automatically updates payload with `expectedDocumentNumber` and retries submission (max 1 retry attempt).